



School of Public Health

Biostatistics for Bachelor of Pharmacy students

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Course outline

Course Description:

- ❑ This module will introduce students about:
 - ❖ principles and concept of Biostatistics
 - ❖ definition of biostatistics and the importance of statistics
 - ❖ basic know-how of managing health and health related
- ❑ Health service of any kind is delivered based on concrete and proven evidences—not by intuitions, anecdotes and preferences
- ❑ This course is the tool that furnishes the learner with the capability of generating credible and dependable evidence to guide health sector activities

Course learning outcome

- 1: Discuss the role of statistics in health sciences and explain the main uses of statistical methods in the broad field of health care.
- 2: Differentiate the various forms of data and variables.
- 3: Describe and use methods of data collection, data organization and presentation, and data summarization.
- 4: Recognize the relative merits and demerits of the different ways of data presentation.
- 5: Calculate and interpret various measures of central tendency and dispersion.

Course learning outcome ...

- 6: Identify and make use of data from existing health records.
- 7: Differentiate the different sources of demographic information.
- 8: Apply different techniques of sampling.
- 9: Carryout statistical estimation and sample size estimation.
- 10: Determine and explain confidence intervals.
- 11: Carryout hypothesis testing and explain the meaning of statistical significance.
- 12: Describe the basic concept of statistical software and its application

Contents

Chapter One : Introduction to Statistics

- Introduction
- Rationale of studying Statistics
- Scales of measurement

Chapter Two: Methods of Data collection, Organization and presentation

- Introduction
- Data collection methods
- Choosing a method of data collection
- Types of questions
- Steps in designing a questionnaire
- Methods of data organization and presentation

Contents...

Chapter Three : Summarizing data

- Measures of Central Tendency
- Measures of Variation

Chapter Four : Demographic Methods and Health Services Statistics

- Sources of demographic data
- Stages in demographic transition
- Vital Statistics
- Measures of Fertility
- Measures of Mortality
- Population growth and Projection
- Health services statistics

Contents ...

Chapter Five : Elementary Probability and probability distribution

- Mutually exclusive events and the additive law
- Conditional Probability and the multiplicative law
- Random variables and probability distributions

Chapter Six : Sampling methods

- Common terms used in sampling
- Sampling methods
- Errors in Sampling

Contents...

Chapter seven : Estimation

- Point estimation.
- Intervalle estimation
- Sampling distribution of means and proportion
- Sample size estimation for cross-sectional data

Contents ...

Chapter Eight : Hypothesis Testing

- The Null and Alternative Hypotheses
- Tests of significance on means and proportions on one and two tailed test
- Test a hypothesis for single population mean and proportion
- Test a hypothesis for difference between two population means and proportions
- Comparing the means of small samples
- Test of significance using the Chi-square and Fisher's exact tests

Contents ...

Chapter Nine: Correlation and Regression

- Correlation analysis
- Regression analysis

Chapter 10: introduction to statistical software's

- Kobo-collect
- Epi-data
- SPSS

Teaching Methods

- ❖ Interactive lecture
- ❖ Small group discussion
- ❖ Large group discussion
- ❖ Seminar presentation
- ❖ Project presentation

Assessment methods

ASSESSMENT TYPE FOR SECOND YEAR ONLY:

- ❖ Continuous assessment accounts: 20%
- ❖ Mid exam accounts: 30%
- ❖ Comprehensive final exam accounts: 50%

- ***Chapter one : Introduction to Biostatistics***

Objectives:

At the end of this section the student should be able to:-

- Define statistics and biostatistics
- Discuss the role of statistics in health sciences
- Explain types of statistics
- Describe common terms in biostatistics
- Describe the importance of statistics in health sciences
- Define and identify the different types of data

Brain storming

- ☐ what is statistics?
- ☐ Is it important to all professionals?
- ☐ Have you heard about vital statistics?
- ☐ Is a statistics a tool or science or art?
- ☐ What is biostatistics?

Definition of statistics

- It is a methodology that scientists and mathematicians have developed for collecting, analyzing, interpreting data and drawing conclusions from information
- Statistics is a **way of thinking about numbers** (collection, analysis, and presentation) with emphasis on relating their interpretation and meaning to the manner in which they are collected

Cont'd...

- The term statistics is used to mean either **statistical data or statistical methods**.
- **Statistical data:** numerical descriptions of things.
 - These descriptions may take form of **counts or measurements**.
- NB:** Even though statistical data always denote figures (numerical descriptions) it must be remembered that **all 'numerical descriptions' are not statistical data**.
- In order say numerical descriptions called **statistics** they must possess the following characteristics:

Characteristics of statistical data

- i. Must be in aggregates
- ii. Must be affected to a marked extent by a multiplicity of causes.
- iii. They must be **enumerated** or **estimated** according to a reasonable standard of accuracy.
 - If the basis happens to be **incorrect** the results are bound to be **misleading**.
- iv. They must have been collected in a **systematic** manner
- v. They must be placed in **relation to each other**

- It refers to a body of methods that are used for **collecting, organising, analysing and interpreting** numerical data for understanding a phenomenon or making wise decisions.
- It is a branch of scientific method and helps us to know in a better way the object under study.

Statistical cycle

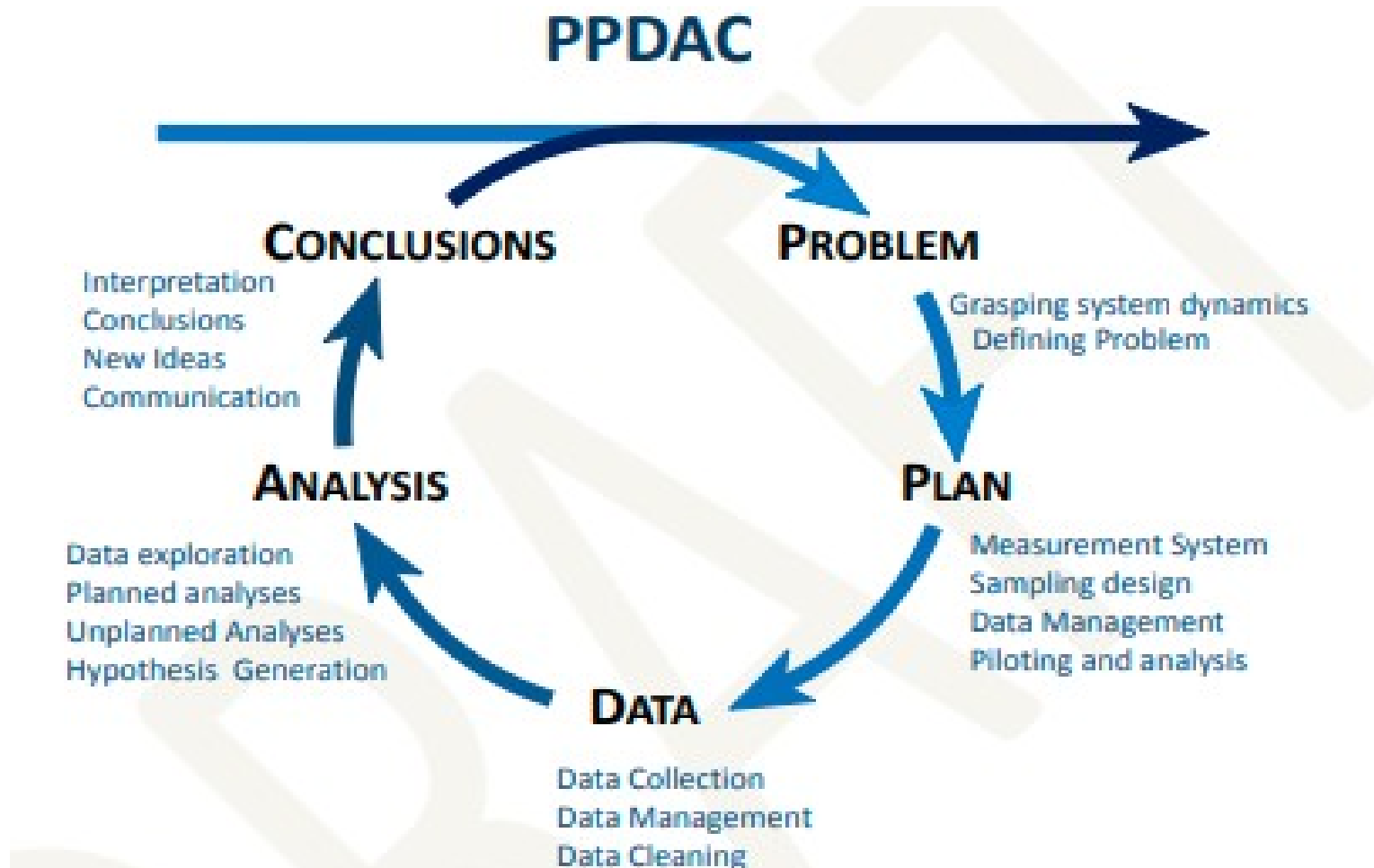


Fig. 1.1: The statistical inquiry cycle

Biostatistics

- When the data analyzed are derived from the **biological sciences** and **health science**,
- We use the term biostatistics to distinguish this particular application of statistical tools and concepts.
- **Biostatistics** is a branch of applied statistics concerned with the application of statistical methods to medical and biological problem.

Rationale of Biostatistics

- ◆ Statistical method is used to explain and predict some of the health outcomes and the direction of epidemics and pandemics
- ◆ It definitely influences decision-makers in public health.
- ◆ Guide public health and other healthcare practitioners on how to go about controlling these diseases.
- ◆ Biostatistics is a critical and invaluable tool in developing public health policy and initiatives
- ◆ Scientific community rely on accurate and timely data to deal with outbreaks of infectious diseases, such as Cholera and COVID-19.

Limitation of statistics

- It deals with only those subjects of inquiry that are capable of being quantitatively measured and numerically expressed
- It deals on aggregates of facts and no importance is attached to individual items- Suited only if their group characteristics are desired to be studied
- Statistical data are only approximate and not mathematically correct

Types of applied statistics

Depending on how data are used, statistics divided into two

1. Descriptive statistics
2. Inferential statistics

1. Descriptive statistics

- All about a set of information that has been collected only
- Consists of the collection, organization, summarization, and presentation of data.
- Focus on just mere descriptive, do not involve generalizing beyond the data at hand.

Types of statistics...

2. Inferential statistics.

- Generalizing from our data to another set of cases or from samples to populations
- Determining relationships among variables, and making predictions
- Uses probability, i.e., the chance of an event occurring

It has two main uses:

- ✓ Making estimates about populations
- ✓ Testing hypotheses to draw conclusions about populations

Definition of common statistical terms

Data : Numbers or non-numbers (i.e. qualities or attributes of something) which contain information

- The raw material of statistics
- Derived/resulted from either measurement or count
- Thus the purpose of statistics is to investigate and evaluate the nature and meaning of these data

Information: Is data that have been processed, organized and has meaning to the user

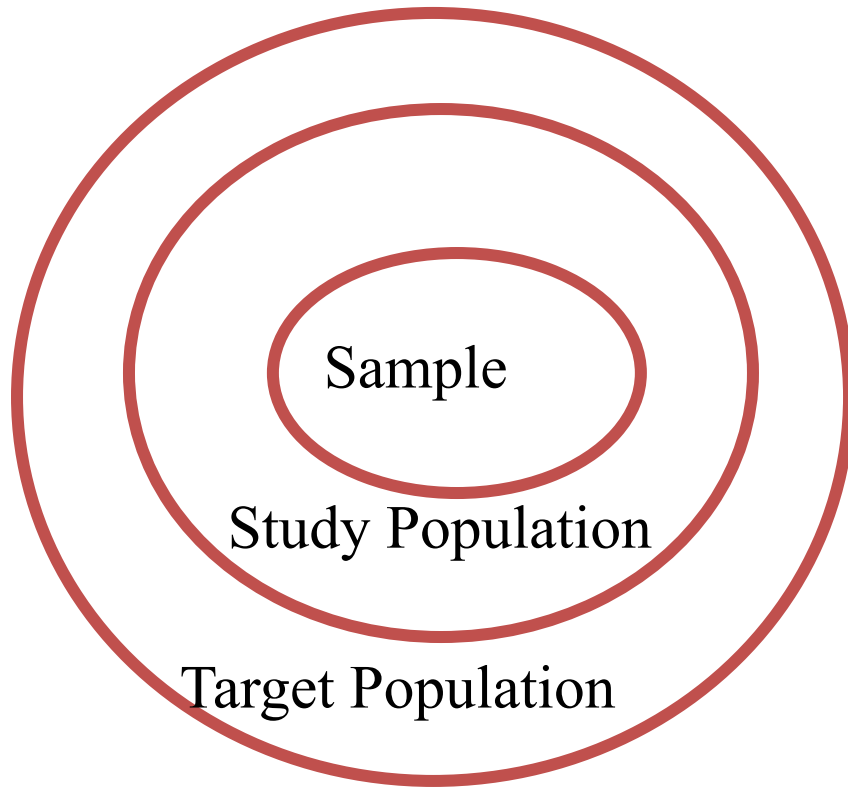
Definition of common...

Population & Sample

Target population: A collection of items that have something in common for which we wish to draw conclusions at a particular time.

Study Population: Part of source population in which the sample is drawn(has a chance of being included in the actual sample)

Sample: A subset of a study population, about which information is actually obtained



E.g.: In a study of the prevalence of depression among adolescents in Addis Ababa, a random sample of sub-cities; Yeka sub-city and other 3-sub-cities of Addis Ababa Town were included.

Target Population: All adolescents in Addis Ababa

Study population: All adolescents in selected sub-cities

Sample: Adolescents selected in each four sub-cities

Definition of common ...

Parameter: It is numerical expression of population measurements

Example

- Population mean (μ)
- Population variance
- Population standard deviation, etc.

Statistic: It is numerical expression of sample measurements.

Example:

- Sample mean
- Sample variance (s^2) and
- Standard deviation (s)

Definition of common ...

- **Variable:** a characteristic that takes different values in different persons, places, or things /not the same when observed in different possessors
 - ✓ E.g. Blood pressure, Heart rate, Height, Weights, Ages, Sex, Temperature, etc.
- **Can be classified as**
 - *Categorical (Qualitative) and*
 - *Numeric variables (Quantitative)*
- **Quantitative Variables** - is a quantifiable characteristic whose values are numbers

Definition of common...

- Numeric variables may be either continuous or discrete.
 - E.g. heights, weights, and body temperatures etc.
- ***Discrete variables***
 - Can be assigned values such as 0, 1, 2, 3 and are said to be countable.
- ***Continuous variables***
 - Assume an infinite number of values in an interval between any two specific values.
 - Obtained by measuring
 - Often include fractions and decimals

Definition of common...

- **Categorical variables**

- Characteristic that can't be quantifiable.
- Variables that can be placed into distinct categories
- Can be either nominal or ordinal.
- E.g. gender (male or female), religion, residence etc.

- **Random variables**: variables whose values are determined by chance

- **Data set**: a collection of data values

- **Data value or a datum** - each value in the data set

Definition of common...

Independent variable: a variable that you believe might influence your outcome measure.

- An independent variable is a hypothesized cause or influence on a dependent variable

Dependent variable: in a research study , the variable that you believe might be influenced or modified by some treatment or exposure.

- It may also represent the variable you are trying to predict.
- Some times called as outcome variable

Measurement/observations

- Assignment of numbers to objects or events according to a set of rules
- ❖ **Variables can be classified in to four basic scales of measurement)** under these broader nonmetric & metric measurements
 - Nominal Scale
 - Ordinal scale
 - Interval scale
 - Ratio scale
- Nominal and ordinal scales are nonmetric measurement scales.

Measurement Scale....

Nominal scale: Categorical data and numbers that are simply used as identifiers or names represent a nominal scale of measurement this consists of two or more categories/classes that are mutually exclusive.

- Nominal measurements have no meaningful rank order among values.

e.g. sex, marital status, blood type, ethnicity, religion, eye color

Measurement Scale....

Ordinal scale: putting variables in graded order

- ✓ You can rank ordinal data but you cannot quantify differences between two ordinal values

- ✓ This has additional quality that the categories are ranked.

e.g. social class, income status, patient status, level of education, smoking status ,cancer stage

- ✓ We try to rank from the lowest to the highest

Measurement Scale....

Interval scale: assigns each measurement to unlimited categories that are equally spaced

- ❑ In addition zero does not represent the absolute lower value

e.g. temperature

- ❑ Not that zero degree Celsius is arbitrary, so it does not make sense to say that 20 degree Celsius is twice hot as 10 degree Celsius

Measurement Scale....

Ratio scale: measurements where both a zero value and distances between different measurements are defined , provide the greatest flexibility in statistical methods that can be used for analyzing data.

➤ In this case zero shows the absence of quantity measured(true zero point).

e.g. height, weight, length etc

Exercise

- Identify the type of data represented by each of the following variables and confirm your answer by giving your own explanation?
- ❖ Blood group, temperature, number of heart attacks, diagnosis of disease, IQ, number of cases of each reportable disease reported by health officer e.g. polio, stage of cancer, the weight of one year old dog(with special diet supplement)was 950gm last month, number of accidents in three year period

Thank you !!